SST 402 — DECISION ANALYSIS

Module 1: Overview and Lesson Plans (Lessons 1–11)

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Kenyatta University — Digital School of Virtual and Open Learning, in collaboration with the School of Pure and Applied Sciences, Department of Mathematics and Actuarial Science. Written by Dr. Christopher Ouma Onyango. The title page of the source bears the code ***SST 402: Decision Analysis****.*

This edition contains all 43 pages of the module. The downloaded page was badly broken: every page carried three streams of text on top of each other — the lesson prose, the two-column iLearn activity plan ("Spark / Individual contribution / Dialogue begins / E-moderator interventions / Time and schedule / Next") and the tables, whose cells had lost their grid and were strung out in cell order (for example the whole week-by-week module plan arrived as "DECISION MAKING UNDER UNCERTAINTY WEEK 2 WEEK 5 OPPORTUNITY LOSS WEEK 9 …"). Nothing was dropped: the tables have been rebuilt cell by cell, the activity plan of each lesson is set out as a box, and every figure and worked answer of the module is reproduced. Where the module's own arithmetic does not reconcile, the number is printed as it stands and the corrected value is given in a shaded note, with the full list in Appendix B.

MODULE PLAN

Week	Topic	Lesson
1	Decision making process	Lesson 1
2	Decision making under uncertainty	Lesson 2
3	Decision making under regret	Lesson 3
4	Decision making under risk	Lesson 4
5	Opportunity loss	Lesson 5
6	Expected value under perfect information	Lesson 6
7	Relationship between EOL, EMV and EVPI	Lesson 7
8	Return to risk ratio	Lesson 8
9	Bayes' decision analysis	Lesson 9
10	Expected utility criterion	Lesson 10
11	Decision trees	Lesson 11
12 & 13	Examination	—

LESSON 1: DECISION MAKING ENVIRONMENT

1.1 Introduction

In any organization the main function of executives is to make decisions such as: (i) should a new product be introduced in the market, (ii) how many units of the product should be introduced, (iii) at what cost and at what price should the product be valued?

A decision problem is therefore how to select the best of the available alternatives or actions, the possible events (states or outcomes of a random process), the probabilities of these events and the criteria (decision rule) according to which the best alternative is selected.

1.2 Lesson learning outcomes

At the end of this lesson you will be able to discuss a decision-making environment.

1.2.1 Decision-making environment

In each of such decision-making problems there are certain common elements. These are:

(a) Alternative courses of action. This is the course of decision making which involves the selection of a single act or action from among some sets of alternative actions.

(b) Uncertainty. In all decision-making problems, uncertainty is found to be a common element. When the outcome of some action is not known in advance the outcome is said to be uncertain. When there are many possible outcomes of events — also known as states of nature — one cannot predict with certainty what will happen.

A decision problem is therefore how to select the best of the available alternatives or actions, the possible events (states or outcomes of a random process), the probabilities of these events and the criteria (decision rule) according to which the best alternative is selected.

Decision analysis falls under two categories:

1. decision making under uncertainty;
2. decision making under risk.

Example. Vincent is a credit manager for the Fine Fabrics Limited. He is currently faced with the question of whether to extend $\$100,000$ credit to a potential new customer, a dress manufacturer. Vincent has three categories for the credit worthiness of a company — poor risk, average risk and good risk — but he does not know which category fits this potential customer.

Experience indicates that 20% of companies similar to this dress manufacturer are poor risks, 50% are average risks and 30% are good risks. If credit is extended the expected credit for poor risks is $-\$15,000$, for average risks $\$10,000$ and for good risks $\$20,000$. If the credit is not extended, the dress manufacturer will turn to another mill. Vincent is able to consult a credit rating organization for a fee of $\$5,000$ per company evaluated. For companies whose actual credit record with the mill turns out to fall into each of the three categories, the table below shows the percentages that were given each of the three possible credit evaluations by the credit rating organizations.

Table 1.1 — Reliability of the credit rating organization

Credit evaluation	Poor risk	Average risk	Good risk
Poor	50%	40%	20%
Average	40%	50%	40%
Good	10%	10%	40%

Develop a decision analysis formulation of this problem by identifying the decision alternatives, the states of nature and payoff table when the credit-rating organization is not used.

Solution. The decision alternatives are *extend credit* and *do not extend credit*; the states of nature are the actual credit record of the customer (poor, average, good). The payoff table is:

Alternative	Poor risk	Average risk	Good risk
Extend credit	−$15,000	$10,000	$20,000
Do not extend credit	$0	$0	$0
Prior probabilities	0.2	0.5	0.3

Activity 1.3 — Decision making process

- **Title:** Decision making process
- **Purpose:** To develop a decision table
- **Brief summary of overall task:** Compare your results with those of your classmates posted on the forum.
- **Individual contribution:** Suppose you are a manager in a restaurant interested in the introduction of a new product. The new product may be introduced either gradually or all at once. (i) Describe the decision-making process for this problem. (ii) Prepare a decision table for this problem.
- **Spark:** Watch *decision making process*.
- **Dialogue begins:** Post your results for the above work out on the forum.
- **E-moderator interventions:** Give feedback on work done by students; coordinate the students' discussions.
- **Time and schedule:** Week 1, for 3 hours.
- **Next:** Decision making under uncertainty.

1.4 Exercise

1.4.1 A small business company valued at $40,000 is offered fire insurance at an annual premium of $500. Suppose that there are just two mutually exclusive outcomes — complete damage and no damage from fire — and that the probability of damage from fire is 0.0001. Identify the courses of action and the alternatives, hence construct the business owner's payoff table for the decision of whether or not to buy fire insurance.

Solution guide.

Acts	Fire	No fire
Buy policy	−$500	−$500
No policy	−$40,000	$0

> *[The module marks this exercise "complete the table"; the completed table is given above, using the value of the business ($40,000) as the loss under "no policy, fire".]*

LESSON 2: DECISION MAKING UNDER UNCERTAINTY

2.1 Introduction

Here the decision maker has absolutely no knowledge, not even the likelihood of an occurrence for any state of nature. In such situations the decision-making behaviour is purely based on his or her attitude towards the unknown. These behaviours are (i) optimism, (ii) pessimism, (iii) least regret.

For instance, consider an idea about a glass of water: an optimist will say "the glass is half full" while the pessimist will say "the glass is half empty". We note that both the pessimist and the optimist are right — the optimist sees an opportunity in every problem while the pessimist sees a problem in every opportunity — but both contribute to society.

2.2 Lesson learning outcomes

At the end of this lesson you will be able to determine the best decision under

1. pessimism or conservatism;
2. optimism or aggression;
3. the coefficient of optimism (Hurwicz's method).

Example. The following table presents a decision-making problem. The states of nature are the states of the economy during one year. The problem is to decide what action to take among three possible courses of action, with the rates of return indicated in Table 2.2.1.

Table 2.2.1 — Rates of return (%) on three courses of action

Courses of action (strategy)	High growth	Medium growth	No change	Low growth
Bonds (B)	12	8	7	3
Stock (S)	15	5	5	−2
Deposits (D)	7	7	7	7

Determine the best decision under (a) pessimism, (b) optimism, (c) the coefficient of optimism (Hurwicz's method).

Solution.

(a) Decision making under pessimism. This is also called the maximin decision process.

Steps: (i) write the minimum value in each action row; (ii) choose the value which is maximum and perform that action.

Strategy	Minimum return
Bonds	3
Stock	−2
Deposits	7

Since the return on deposit is the highest among the minimum returns, the investor should invest in deposits.

(b) Decision making under optimism (maximax).

Steps: (i) write the maximum number in each action row; (ii) choose the maximum number and take that action.

Strategy	Maximum return
Bonds	12
Stock	15

Strategy	Maximum return
Deposits	7

Under optimism the best decision is to invest in stock.

(c) Decision making under the coefficient of optimism (Hurwicz's method). This is a case where the decision maker is neither too optimistic nor pessimistic.

Steps: (i) choose α between 0 and 1, where 1 means optimistic and 0 means pessimistic; (ii) choose the largest and the smallest number of each action; (iii) multiply the largest payoff (row-wise) by α and the smallest by $(1-\alpha)$; (iv) pick the action with the largest sum.

Suppose in the above case $\alpha = 0.6$. The total returns are then

$$\text{Bonds: } 12(0.6) + 3(0.4) = 8.4,$$
$$\text{Stock: } 15(0.6) + (-2)(0.4) = 8.2,$$
$$\text{Deposits: } 7(0.6) + 7(0.4) = 7.0.$$

In this case the best decision is to invest in bonds.

Activity 2.3 — Decision making under uncertainty

- **Title:** Decision making under uncertainty
- **Purpose:** To determine the best option under uncertainty
- **Brief summary of overall task:** Read on decision making under uncertainty.
- **Individual contribution:** Janet has some money to invest in one of four options A, B, C, D. The return on investment depends on the quality of items produced. The possible returns of each strategy and state of nature are given in the table below. Determine the decision under (a) pessimism, (b) optimism, (c) Hurwicz's method with $\alpha = 0.7$.

Course of action (strategy)	Grade 1	Grade 2	Grade 3
A	600	700	750
B	500	650	550
C	450	290	420
D	300	420	620

- **Spark:** Watch the video on decision making under uncertainty.
- **Dialogue begins:** Post your workings.
- **E-moderator interventions:** Interpret the solutions obtained in the work on the individual contribution above; coordinate discussions on the forum; provide feedback on the assignment.
- **Time and schedule:** Week 2, for 3 hours.
- **Next:** Decision making under regret.

2.4 Assessment

Question 2.4.1. A and J Company Limited has two possible alternatives under consideration: drill or sell the land. The possible states of nature are that the land contains oil and that it does not, as shown in the table below. Since the consulting geologist has estimated that there is one chance in four of oil (and so three chances in four of no oil), the prior probabilities of the two states are 0.25 and 0.75 respectively.

Alternatives	Oil	Dry
Drill for oil	700	−100
Sell the land	90	90
Prior probability	0.25	0.75

Determine the best decision under (i) the minimax criterion, (ii) the maximin criterion.

Solution guide. (i) Sell the land (minimum value is 90). (ii) The maximin criterion is to sell the land.

Question 2.4.2. Larry is considering producing a novelty item that would be sold through subsidiary shops. Larry has decided on a selling price of $3.50 for the item. The item's variable cost of production is $2 per unit, with a fixed cost of $0.50. Larry has marketed his product to the shops and believes that the demand for the item will be 2,000 units, 3,000 units, 4,000 units or 5,000 units.

a. Set up the payoff table for Larry's decision.

b. Determine the number of units that Larry should produce using each of the following criteria: (i) maximax, (ii) maximin.

LESSON 3: DECISION MAKING UNDER REGRET

3.1 Introduction

Regret amount is the amount foregone for not taking the optimal decision. This method is also known as the minimax regret criterion. It is a payoff of what could have been the best decision in the circumstance minus the payoff of the actual decision in the circumstances.

3.2 Lesson learning outcomes

At the end of this lesson you will be able to determine the best decision under the minimum regret criterion in decision analysis.

3.2.1 Decision making under regret

The first step is to set up a regret table.

Step 1. Take the largest number in each state-of-nature column. *Step 2.* Subtract all the elements in that state of nature from it. *Step 3.* Choose the maximum number of each action. *Step 4.* Choose the minimum number from step (iii) and take that action.

Example. Refer to Table 2.2.1 in Lesson 2 and determine the best decision under regret.

Solution. From Table 2.2.1 we develop the regret table below; the column maxima of Table 2.2.1 are 15 (high growth), 8 (medium growth), 7 (no change) and 7 (low growth).

Table 3.2.1 — Regret (opportunity loss) table

Course of action (strategy)	High growth	Medium growth	No change	Low growth	Maximum
Stock (S)	0	3	2	9	9
Bonds (B)	3	0	0	4	4
Deposit (D)	8	1	0	0	8

In this case the best decision is to invest in bonds, since it minimizes the regret.

Activity 3.3 — Decision making under regret

- **Title:** Decision making under regret
- **Purpose:** To determine decision making under minimum regret
- **Brief summary of overall task:** Watch *decision making under regret*.
- **Individual contribution:** A telecommunication company is considering three options for managing its data processing operation: continue with its own staff, outsource to a vendor, or use a combination of its own staff and an outside vendor. The cost of the operation depends on future demand. The annual cost of each option, in thousands of Kenya shillings, depends on demand as follows. Construct the corresponding regret table, hence determine the best decision under regret.

Staffing options	d_1**: High**	d_2**: Average**	d_3**: Low**
Own staff	650	900	800
Outsource	650	600	650
Combination	600	300	500

- **Spark:** —
- **Dialogue begins:** Post your workings on forum 3.
- **E-moderator interventions:** Coordinate the discussions on the forum; note any differences in solutions obtained and improve on your workings; provide feedback.
- **Time and schedule:** Week 3, for 3 hours.
- **Next:** Decision making under risk.

3.4 Assessment

3.4.1 A telecommunications company is considering the same three options as in Activity 3.3. Required: construct the corresponding opportunity loss table.

Solution guide — regret table (as printed in the module):

Staffing options	**High**	**Average**	**Low**
Own staff	0	0	50
Outsource	0	300	600
Combination	0	150	300

> *[The costs in Activity 3.3 are a **cost** table, so the opportunity loss of an act is its cost minus the smallest cost in the column, not the largest minus the cost. Computed that way the table is:*
>
> | ***Staffing options*** | ***High*** | ***Average*** | ***Low*** | ***Maximum*** |
> |---|---|---|---|---|
> | *Own staff* | *50* | *600* | *300* | *600* |
> | *Outsource* | *50* | *300* | *150* | *300* |
> | *Combination* | *0* | *0* | *0* | *0* |
>
> *and the minimax-regret decision is **combination** (regret 0). The module's table is the "largest minus element" form, which is the payoff-table rule, and it would point to own staff.]*

3.4.2 Larry is considering producing a novelty item that would be sold through subsidiary shops (see Question 2.4.2). Set up the regret table and determine the number of units that Larry should produce using minimax regret.

LESSON 4: DECISION MAKING UNDER RISK

4.1 Introduction

Risk implies the degree of uncertainty and the inability to fully control the outcomes or consequences of one's actions. Risk, or the elimination of risk, is an effort that managers employ. The decision maker evaluates the decision strategies prior to making any decision.

4.2 Lesson learning outcomes

At the end of this lesson you will be able to

1. explain the steps in the decision-making process under risk;
2. evaluate the expected monetary value (EMV) of a decision problem.

4.2.1 Steps in decision making under risk

1. The problem is defined and all feasible alternatives are considered. The possible outcomes for each alternative are evaluated.
2. Outcomes are discussed on the basis of their monetary payoffs or net gains in reference to assets or time.
3. Various uncertainties are quantified in terms of probabilities.
4. The quality of the optimum strategy depends upon the quality of the judgements.

4.2.2 Expected monetary value (EMV)

Steps: (i) for each action, multiply the probability and the payoff; (ii) add up the results by row; (iii) choose the largest number and take that action.

Example 1. Consider the following probabilities for the different economic growth rates, based on the table below. Determine the best decision strategy using the expected monetary value criterion.

Courses of action (strategy)	High growth	Medium growth	No change	Low growth
Bonds (B)	12	8	7	3
Stock (S)	15	5	5	−2
Deposit (D)	7	7	7	7
Probabilities	0.4	0.3	0.2	0.1

Solution. First calculate the EMV for each strategy:

$$E(X) = \sum_{i=1}^{n} x_i \, p\left(x_i\right).$$

Bonds: $12(0.4) + 8(0.3) + 7(0.2) + 3(0.1) = 8.9$
Stocks: $15(0.4) + 5(0.3) + 5(0.2) + (-2)(0.1) = 8.3$
Deposits: $7(0.4) + 7(0.3) + 7(0.2) + 7(0.1) = 7.0$

Conclusion. The best decision is to invest in bonds, since it has the highest EMV.

> *[The module prints the stock row of the payoff table as "15 5 5 –2" and the bonds row as "12 8 8 3"; the EMV line that follows (8.9) requires the bonds row 12, 8, 7, 3, and that is what is used here.]*

Example 2. A business person at Muthurwa market sells vegetables. She receives her supplies every Thursday morning, and this consignment must last until the following Thursday, when the new consignment is received. Any vegetables left unsold next Thursday are destroyed. The vegetables sell at $10 per bag and cost $8 per bag. The weekly demand for vegetables at her grocery varies from week to week as shown below from past experience.

Bags demanded	10	11	12	13	14
Probability	0.1	0.2	0.4	0.2	0.1

How many bags of vegetables should she stock every Thursday?

Solution. The profit function: let D be the quantity ordered and S the quantity sold. The profit per bag sold is $\$10 - \$8 = \$2$, hence

$$\text{profit} = \begin{cases} 2D, & D \le S, \\ 10S - 8D, & S < D. \end{cases}$$

Payoff table (rows: bags stocked; columns: demand)

Stocked	10	11	12	13	14	EMV
10	20	20	20	20	20	20.0
11	12	22	22	22	22	21.0
12	4	14	24	24	24	20.0
13	−4	6	16	26	26	15.0
14	−12	−2	8	18	28	8.0

The expected payoffs are the row products weighted by the demand probabilities, for example for 11 bags

$$0.1(12) + 0.2(22) + 0.4(22) + 0.2(22) + 0.1(22) = 1.2 + 4.4 + 8.8 + 4.4 + 2.2 = 21.0.$$

Conclusion. The best decision is the one that maximizes EMV; thus the best decision is to stock 11 bags every Thursday.

> *[The module prints the EMV row as "20 21 20 15 8.6". The last entry recomputes as* $0.1(-12) + 0.2(-2) + 0.4(8) + 0.2(18) + 0.1(28) = 8.0$. *The decision (11 bags) is unaffected.]*

Activity 4.3 — Decision making under risk

- **Title:** Decision making under risk
- **Purpose:** To determine the best decision under the expected monetary value (EMV).
- **Brief summary of overall task:** Find how many dozen roses should be purchased for the next Valentine's Day sales, and the associated expected profit.
- **Individual contribution (case analysis):** The Special Occasions Company, a nationwide distributor of flowers and greeting cards, has just contracted with Global Floral to purchase up to 500,000 dozen red and yellow roses for sale in the week before Valentine's Day. Special Occasions supplies flower shops and other retail outlets from regional distribution centres throughout the country. The problem now facing the company is how many dozen roses to buy. If the company doesn't buy enough, it loses potential business; if it buys more than are demanded, it loses the costs of purchasing and shipping these unsold roses. Any unsold roses are given to local hospitals and retirement homes or are disposed of at the area landfill. The contract calls for a base payment of \$150, 000 plus \$5 for each dozen roses bought by Special Occasions. The company's accounting department has determined that the cost of shipping the roses from Global Floral's greenhouses averages \$1.50 per dozen. The marketing department has estimated the demand (in dozens) for roses, and its probability distribution, as follows. Assuming the roses sell for an average of \$18 per dozen, set up the payoff table and use the expected value criterion to determine the best decision.

Demand (thousands of dozens)	100	150	200	225	250
Probability	0.10	0.40	0.20	0.20	0.10

- **Spark:** Watch *decision making under risk*.
- **Dialogue begins:** Post your workings on forum 3 (as printed in the module).
- **E-moderator interventions:** Coordinate the discussions on the forum; provide feedback.
- **Time and schedule:** Week 4, for 3 hours.
- **Next:** Opportunity loss.

4.4 Assessment

Question 4.4.1. The manager of a flower shop promises its customers delivery within four hours on all flower orders. All flowers are purchased on the previous day and delivered by 8.00 a.m. the next morning. The daily demand for roses is as follows.

Dozens of roses	**70**	**80**	**90**	**100**
Probability	0.1	0.2	0.4	0.3

The manager purchases roses for \$10 per dozen and sells them for \$30. All unsold roses are donated to a local hospital. Determine (a) the conditional profit function, (b) the number of dozens of flowers which should be ordered to maximize profits.

Solution guide. (a) Marginal profit $10S - 30D$ written in terms of the demand D and the quantity sold S; (b) the best decision is to purchase 90 dozens, whose EMV is 168.

> *[With a profit of* \$20 *per dozen sold and a loss of* \$10 *per unsold dozen the expected profits are 120, 150, 168 and 168 (in the module's units, which correspond to* \$2 *gain and* \$1 *loss per dozen); the module's answer, 90 dozens, is correct.]*

Question 4.4.2. Research has shown that in Nairobi, Kenya more money is being spent on restaurant food than is being spent buying groceries. Many grocery chains, in an effort to keep up with this trend, have added extensive takeout sections. Jimm Limited is a regional chain that has joined the movement. A popular packaged meal is made up of broiled chicken, potato salad and green salad. However, because of several television news specials about potentially spoiled food, Jimm Ltd has a company policy of giving all unsold food to a local orphanage at the end of each day. Although the company does not mind giving the food away, it would rather sell all the meals it prepares. The manager feels that on any week day the minimum demand for the broiled chicken meal will be 50 and the maximum demand 90.

Based on historical data, the probabilities for each demand level are given below.

Demand	**50**	**60**	**70**	**80**	**90**
Probability	0.05	0.1	0.20	0.40	0.25

The fixed cost of setting up to cook the meals is Ksh 120, the variable cost amounts to Ksh 150 per meal and the selling price is Ksh 470 each. Determine how many meals should be prepared.

LESSON 5: OPPORTUNITY LOSS

5.1 Introduction

Suppose that we view each possible outcome in terms of a measure that expresses the difference between the payoff of an act and the best payoff that could have been achieved. This measure is called an **opportunity loss**. Thus opportunity loss is the amount of payoff that is foregone by not selecting the act that has the maximum payoff for the event that occurs. This is decision making in which the decision maker would like to minimize the loss in taking an action among alternative actions.

5.2 Lesson learning outcomes

At the end of this lesson you will be able to construct an opportunity loss table and determine the best decision under expected opportunity loss.

5.2.1 Steps in constructing an opportunity loss table

Step 1. Set up a loss (payoff) matrix. *Step 2.* For each action, multiply the probability and the loss, then add up for each action. *Step 3.* Choose the action with the smallest expected opportunity loss.

> *[The module numbers these two subsections 5.5.1 and 5.5.2, although they follow §5.2. They are numbered 5.2.1 and 5.2.2 here.]*

5.2.2 Determining the best decision under the opportunity loss criterion

Example. Suppose a toy company has established the payoff table below (Table 5) for manufacturing toys of different quality of gears and levers, action springs, weights and pulleys.

Table 5 — Payoff table (profits in $)

Level of demand (quality)	Probability	Gears & levers	Action springs	Weights & pulleys
Light	0.1	25,000	$-10,000$	$-125,000$
Moderate	0.7	400,000	440,000	400,000
Heavy	0.2	650,000	740,000	750,000

Construct the loss table and compute the expected loss.

Solution — loss table. The column maxima are 25,000, 440,000 and 750,000.

Level of demand (quality)	Probability	Gears & levers	Action springs	Weights & pulleys
Light	0.1	0	35,000	150,000
Moderate	0.7	40,000	0	40,000
Heavy	0.2	100,000	10,000	0
Expected loss		**48,000**	**5,500**	**43,000**

The expected losses are, for example for gears and levers,

$$0.1(0) + 0.7(40,000) + 0.2(100,000) = 0 + 28,000 + 20,000 = 48,000,$$

and for action springs

$$0.1(35,000) + 0.7(0) + 0.2(10,000) = 3,500 + 2,000 = 5,500.$$

Conclusion. The best decision is the one that minimizes the expected opportunity loss — here *action springs*, with an expected loss of 5,500.

Activity 5.3 — Opportunity loss

- **Title:** Opportunity loss
- **Purpose:** To determine the best decision under minimax decision analysis.
- **Brief summary of overall task:** Watch *opportunity loss*.
- **Individual contribution:** A manufacturer of children's toys must determine its advertising budget for the coming year. Realizing that sales depend on the nation's economic situation, the manufacturer constructed the payoff table below. The table shows profits in millions of dollars that the manufacturer expects with high, average and low advertising costs in the event of a severe recession, a mild recession, a weak recovery and a strong recovery. (i) Construct the corresponding opportunity loss table. (ii) Assume that economists predict a 20% chance of a severe recession, a 25% chance of a mild recession, a 40% chance of a weak recovery and a 15% chance of a strong recovery. Which advertising budget should the manufacturer choose if it wishes to minimize expected opportunity loss?

State of nature	High advertising	Average advertising	Low advertising
Severe recession	−10	−6	0
Mild recession	−5	−1	1
Weak recovery	4	3	2
Strong recovery	12	9	4

- **Dialogue begins:** Compare your results with those posted by your classmates; note any differences in the approaches used by other students.
- **E-moderator interventions:** Coordinate the discussions on the forum; provide feedback.
- **Time and schedule:** Week 5, for 3 hours.
- **Next:** Expected value under perfect information.

5.4 Assessment

5.4.1 A garment manufacturer wishes to select one of three feasible prototype designs for a new, longer-wearing garment. The costs of making the garments are given below.

Prototype	Fixed cost ($)	Variable cost per unit ($)
A	60,000	30
B	90,000	20
C	120,000	15

There are three levels of unit sales: 4,000 units, 7,000 units and 10,000 units; the respective probabilities are 0.30, 0.50 and 0.20. The selling price will be $75 per garment.

i. Develop the payoff table using total profit as the payoff measure.

ii. Construct the expected opportunity loss table and determine the expected opportunity loss.

Solution guide. With a selling price of $75 and production equal to sales, the profit of a prototype is $(75 - v)D - F$, where D is demand, v the variable cost and F the fixed cost.

(i) Payoff table

Sales event	Probability	A	B	C
4,000	0.30	120,000	130,000	120,000
7,000	0.50	255,000	295,000	300,000
10,000	0.20	390,000	460,000	480,000
Expected payoff		**241,500**	**278,500**	**282,000**

(ii) Opportunity loss table

Sales event	Probability	A	B	C
4,000	0.30	10,000	0	10,000
7,000	0.50	45,000	5,000	0
10,000	0.20	90,000	20,000	0
Expected opportunity loss		**43,500**	**6,500**	**3,000**

Purchasing garment C is the best act, and option C is optimal.

> *[The module prints only the columns that its solution fills in (A and B in the payoff table, A and B in the loss table). The values above complete the table; they agree with the module's expected losses of 43,500, 6,500 and 3,000.]*

LESSON 6: EXPECTED VALUE UNDER PERFECT INFORMATION

6.1 Introduction

The decision on expected opportunity loss has been determined without the benefit of any information except that acquired through experience with other toys, but it is possible to secure better information about next season's demand by test marketing, by taking opinion and attitude surveys, or by obtaining inside information concerning a competitor's plans. We need to determine how much the decision maker will be willing to pay for additional information. The EVPI helps to determine the worth of an insider who owns perfect information.

6.2 Lesson learning outcomes

At the end of this lesson you will be able to explain the steps in determining EVPI, and determine the EVPI of a decision problem and interpret its meaning.

6.2.1 Steps in determining EVPI

Step 1. Take the maximum payoff for each state of nature. *Step 2.* Multiply each case by the probability for that state of nature, then add them up. *Step 3.* Subtract the expected payoff with no information from the expected payoff under certainty:

$$\text{EVPI} = \text{expected payoff under certainty} - \text{maximum expected payoff with no information.}$$

Example. Use Table 5 in Lesson 5 to evaluate the EVPI for the problem.

Solution.

(a) Expected payoff under uncertainty (EMV of each act):

Act	Light (0.1)	Moderate (0.7)	Heavy (0.2)	Expected profit
Gears & levers	2,500	280,000	130,000	412,500
Action springs	$-1,000$	308,000	148,000	455,000
Weights & pulleys	$-12,500$	280,000	150,000	417,500

(b) Expected payoff under certainty (maximum payoff in each state, weighted):

Level of demand	Probability	Max payoff	Max payoff × probability
Light	0.1	25,000	2,500
Moderate	0.7	440,000	308,000
Heavy	0.2	750,000	150,000
Expected payoff under certainty			**460,500**

The best decision is to manufacture action springs, with an expected payoff with no information of $\$455,000$. Now

$$\text{EVPI} = 460,500 - 455,000 = 5,500.$$

> *[The module prints the expected payoff under certainty as 460,000 while its own three products add to 460,500, and it prints the gears-and-levers expected profit as 421,500 (Lesson 7 prints 412,500). The figures 460,500 and 412,500 are used here, since they are the ones consistent with the stated EVPI of 5,500 and with the minimum expected loss of 5,500 found in Lesson 5.]*

6.3 Interpretation

The EVPI represents the greatest amount of money a decision maker will be willing to pay in order to obtain perfect information about what the demand would be. It may also be interpreted as: $\$5,500$ is the increase in the decision maker's expected payoff that may be attributed to perfect knowledge of demand.

Activity 6.3 — Expected value under perfect information

- **Title:** Expected value under perfect information
- **Purpose:** The purpose of EVPI is to determine the worth of an insider who owns perfect information.
- **Brief summary of overall task:** Watch *EVPI* and read the EVPI notes.
- **Individual contribution:** Work out the exercise provided in 6.4.
- **Spark:** —
- **Dialogue begins:** Discuss the steps involved in determining EVPI in decision analysis.
- **E-moderator interventions:** Providing feedback on the posts.
- **Time and schedule:** Week 6, for 3 hours.
- **Next:** Relationship between EOL, EMV and EVPI.

6.4 Exercise

Question 6.4.1. A dairy firm wants to determine the quantity of butter it should produce to meet the demand. Past records have shown the following pattern.

Quantity required (kg)	15	20	25	30	35	40	50
Number of days demand occurred	6	14	20	80	40	30	10

The stock levels are restricted to the range 15 to 50 kg due to inadequate storage facilities. Butter costs Ksh 40 per kg and is sold at Ksh 50 per kg.

a. Write down the conditional profit function.

b. Construct a conditional profit table.

c. Determine the action alternative associated with the maximization of the expected profit.

d. Determine the expected value of perfect information (EVPI).

Solution guide. Let S be the quantity in stock and D the demand. The total number of days is 200, so the demand probabilities are 0.03, 0.07, 0.10, 0.40, 0.20, 0.15 and 0.05.

(a) Conditional profit. Butter costs Ksh 40 per kg and sells at Ksh 50 per kg, so

$$\text{profit} = 50\min(S, D) - 40S.$$

(b) Conditional profit table (rows: stock S; columns: demand D)

Stock (kg)	15	20	25	30	35	40	50
15	150	150	150	150	150	150	150
20	−50	200	200	200	200	200	200
25	−250	0	250	250	250	250	250
30	−450	−200	50	300	300	300	300
35	−650	−400	−150	100	350	350	350
40	−850	−600	−350	−100	150	400	400
50	−1,250	−1,000	−750	−500	−250	0	500

(c) Expected profit table

Possible demand	Probability	15	20	25	30	35	40	50
15	0.03	4.50	−1.50	−7.50	−13.50	−19.50	−25.50	−37.50
20	0.07	10.50	14.00	0	−14.00	−28.00	−42.00	−70.00
25	0.10	15.00	20.00	25.00	5.00	−15.00	−35.00	−75.00
30	0.40	60.00	80.00	100.00	120.00	40.00	−40.00	−200.00

Possible demand	Probability	15	20	25	30	35	40	50
35	0.20	30.00	40.00	50.00	60.00	70.00	30.00	−50.00
40	0.15	22.50	30.00	37.50	45.00	52.50	60.00	0
50	0.05	7.50	10.00	12.50	15.00	17.50	20.00	25.00
EMV (Ksh)		**150**	**192.50**	**217.50**	**217.50**	**117.50**	−32.50	−407.50

(d) Since the maximum EMV is Ksh 217.50, attained for a stock of 25 kg and also of 30 kg, the expected profit with perfect information is

Demand	Probability	Best profit	Probability × best profit
15	0.03	150	4.50
20	0.07	200	14.00
25	0.10	250	25.00
30	0.40	300	120.00
35	0.20	350	70.00
40	0.15	400	60.00
50	0.05	500	25.00
EPPI			**318.50**

$$\text{EVPI} = \text{EPPI} - \text{maximum EMV} = 318.50 - 217.50 = \text{Ksh } 101.$$

LESSON 7: RELATIONSHIP BETWEEN EOL, EMV AND EVPI

7.1 Introduction

In this lesson we investigate the relationship between EMV, EOL and EVPI.

7.2 Lesson learning outcomes

At the end of this lesson you will be able to explain the relationship between EMV, EOL and EVPI in decision analysis.

Example. Refer to Table 5 in Lesson 5. Investigate the relationship between EOL, EMV and EVPI.

Solution. We note that $\$5,500$ — the minimum expected opportunity loss of Lesson 5 — is the same amount as the EVPI of Lesson 6. Thus

$$\text{EVPI} = \text{EOL of the optimal act} = \text{expected payoff under certainty} - \text{maximum EMV}.$$

Table 7.2.1 — The relationship between EMV, EOL and EVPI

Act	EMV	EOL	EVPI
Gears & levers	412,500	48,000	460,500
Action springs	455,000	5,500	460,500
Weights & pulleys	417,500	43,000	460,500

For each act, the EMV and the EOL add up to the expected payoff under certainty:

$$\text{EMV} + \text{EOL} = \text{EPPI}, \qquad \text{EVPI} = \text{EPPI} - \max \text{EMV} = \min \text{EOL}.$$

So, for action springs, $455,000 + 5,500 = 460,500$.

Activity 7.3 — Relationship between EMV, EOL and EVPI

- **Title:** Relationship between EMV, EOL and EVPI
- **Purpose:** To investigate the relationship between EOL, EMV and EVPI.
- **Brief summary of overall task:** Post the workings on forum 7.
- **Individual contribution:** The following is a payoff table. Determine the EMV, EOL and EVPI, hence determine their relationship.

Event	Probability	A1	A2	A3
E1	0.3	10	20	30
E2	0.5	40	−20	20
E3	0.2	20	10	−10

- **Spark:** —
- **Dialogue begins:** Discuss how one would calculate EMV given the EOL and EVPI values; discuss alternative methods for determining the relationship of EMV, EOL and EVPI in decision analysis.
- **E-moderator interventions:** Giving feedback on students' work.
- **Time and schedule:** Week 7, for 3 hours.
- **Next:** Return to risk ratio.

7.4 Assessment

7.4.1 A real estate investor in a city is considering three investments: an apartment building, a strip mall and an office building. Returns from each investment alternative depend on future population growth. The investor has developed three growth scenarios, showing the revenue from each investment alternative along with their probabilities of occurrence, in the table below.

Population growth	Probability	Apartment, A1	Mall, A2	Office, A3
High	0.50	150,000	75,000	100,000
Average	0.30	10,000	30,000	50,000
Slow	0.20	−75, 000	15,000	−200, 000

i. Compute the expected opportunity loss for each act, hence determine the optimal act.

ii. Compute the EMV and EVPI and make deductions.

Solution guide.

Opportunity loss table

Population growth	Probability	A1	A2	A3
High	0.50	0	75,000	50,000
Average	0.30	40,000	20,000	0
Slow	0.20	90,000	0	215,000
Expected opportunity loss		**30,000**	**43,500**	**68,000**

The best act using EOL is A1; you now need to work out the EMV and EVPI and make your deductions.

> *[The module prints the third expected opportunity loss as 65,000; its own three products are* $25,000 + 0 + 43,000 = 68,000$*. The choice of A1 is unaffected.]*

7.4.2 An author is trying to choose between two publishing companies that are competing for the market rights to her new novel. Company A offered \$10, 000 plus \$2 per book. Company B has offered the author \$2, 000 plus \$4 per book sold. The author believes that 5 levels of demand for the book are possible: 1,000, 2,000, 5,000, 15,000 and 50,000 copies.

a. Determine the payoff for each level of demand for companies A and B.

b. Construct the payoff table indicating the events and alternative courses of action.

c. Construct an opportunity loss table.

d. Determine the EVPI.

e. Determine the relationship between EMV, EOL and EVPI from your results.

LESSON 8: RETURN TO RISK RATIO

8.1 Introduction

Neither the EMV nor the EOL take into account the variability of the payoffs of the courses of action under the different events. To take the variability of events into account we compute the variance and the standard deviation of each course of action.

8.2 Lesson learning outcomes

At the end of this lesson you will be able to determine the mean and variance of each course of action, and evaluate the best decision using the return to risk ratio (RTRR).

8.2.1 Mean and variance for a course of action

We first compute the variance of each course of action. The variance of a course of action A is defined by

$$\text{var}(A) = \sum_i p_i \left(x_i - \text{EMV} \right)^2,$$

and consequently the standard deviation is the square root of the variance. From these values we compute the coefficient of variation of each course of action:

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100\%,$$

where σ is the standard deviation.

The larger the coefficient of variation, the more the risk. Consequently we seek, among the actions, the one that gives the least coefficient of variation. However, when there are large differences in the amount of variability between the events, a criterion other than the EMV or the EOL is needed to express the relationship between the return (that is, the EMV) and the risk (as expressed by the standard deviation). The proportion

$$\text{RTRR} = \frac{\text{EMV}}{\sigma}$$

is known as the **return to risk ratio**.

Example. You are a marketing manager for a product company considering the introduction of organic salad dressings. You need to develop a marketing plan and must decide whether you will have a gradual introduction or a concentrated introduction. You estimate that if there is a low demand for the salad dressings your first year's profit will be \$1 million for a gradual introduction and −\$5 million for the concentrated introduction. If there is high demand, you estimate that your first year's profit will be \$4 million for the gradual introduction and \$10 million for the concentrated introduction.

a. Construct a payoff table for these alternative courses of action.
b. Suppose there is a probability of 0.6 that there will be low demand; compute the EMV of each alternative and determine how to market the salad dressing.
c. Determine the coefficient of variation for the two courses of action.
d. Evaluate the RTRR, hence make your conclusions.

Solution.

(a) Payoff table

Event (level of demand)	Probability	Gradual	Concentrated
High	0.4	4	10
Low	0.6	1	−5

(b) EMV table

Event (level of demand)	Probability	Gradual	Concentrated
High	0.4	1.6	4.0
Low	0.6	0.6	−3.0
EMV		**2.2**	**1.0**

The highest EMV is \$2.2 million, so on the EMV criterion the gradual introduction is preferred.

(c) Coefficient of variation

$$\text{var(gradual)} = (4-2.2)^2(0.4) + (1-2.2)^2(0.6) = 1.296 + 0.864 = 2.16,$$

so $\sigma_{\text{gradual}} = \sqrt{2.16} = 1.4697$, and

$$\text{CV}_{\text{gradual}} = \frac{1.4697}{2.2} \times 100\% = 66.8\%.$$

Similarly,

$$\text{var(concentrated)} = (10-1)^2(0.4) + (-5-1)^2(0.6) = 32.4 + 21.6 = 54,$$

so $\sigma_{\text{concentrated}} = \sqrt{54} = 7.348$, and

$$\text{CV}_{\text{concentrated}} = \frac{7.348}{1.0} \times 100\% = 734.8\%.$$

8.2.2 Evaluating the return to risk ratio

In the above example we note that the variability between these two approaches to the introduction of salad dressings is very wide. In such a case it is prudent not only to make a decision at this stage, but also to determine the return to risk ratio:

$$\text{RTRR}_{\text{gradual}} = \frac{2.2}{\sqrt{2.16}} = \frac{2.2}{1.4697} = 1.497, \qquad \text{RTRR}_{\text{concentrated}} = \frac{1}{\sqrt{54}} = \frac{1}{7.348} = 0.136.$$

The gradual introduction earns about 1.50 units of expected return for every unit of risk, against 0.14 for the concentrated introduction: it has both the higher EMV and the smaller coefficient of variation. The best decision is therefore to implement the strategy gradually.

> *[The module's text at this point reads "This suggests that the RTRR for the decision of gradual introduction has much higher risk than the concentrated strategy… Consequently, gradual implementation has a higher EMV than a concentrated implementation. Thus, the best decision is to implement the strategy gradually." The RTRR of the gradual strategy is the **larger** of the two (1.497 against 0.136), which means more return per unit of risk, and its coefficient of variation is the smaller (66.8% against 734.8%); the conclusion drawn by the module — implement gradually — is the one these figures support.]*

Activity 8.3 — Return to risk ratio

- **Title:** Return to risk ratio (RTRR)
- **Purpose:** To determine the action taking into account the variability of events.
- **Brief summary of overall task:** Watch the video on the coefficient of variation and the return to risk ratio.
- **Individual contribution:** A stock has the following predicted returns under the following economic situations. Determine (i) the standard deviation, (ii) the coefficient of variation, (iii) the RTRR.

Economic condition	Probability	Return
Recession	0.3	50
Stable growth	0.3	100
Moderate growth	0.3	120

- **Spark:** —
- **Dialogue begins:** Post your results in discussion forum 8.
- **E-moderator interventions:** Compare your results with those posted by your classmates and note any differences in the solution algorithm; engage learners on the importance of coefficient of variation and RTRR values in decision making.
- **Time and schedule:** Week 8, for 3 hours.
- **Next:** Bayes' decision analysis.

8.4 Assessment

The table below presents predictions of returns for stocks A and B under different economic conditions.

Event (economic conditions)	Probability	A	B
Recession	0.10	30	-50
Stable economy	0.40	70	30
Moderate growth	0.30	100	250
Boom	0.20	150	400

a. Determine the coefficient of variation for each action.

b. Find the return to risk ratio for each action.

c. Compare (i) and (ii) and comment on any differences.

Solution guide.

(a) The expected returns are

$$\text{EMV}_A = 0.1(30) + 0.4(70) + 0.3(100) + 0.2(150) = 91,$$
$$\text{EMV}_B = 0.1(-50) + 0.4(30) + 0.3(250) + 0.2(400) = 162,$$

with standard deviations $\sigma_A = 35.48$ and $\sigma_B = 158.5$, so that

$$\text{CV}_A = \frac{35.48}{91} \times 100\% = 39.1\%, \qquad \text{CV}_B = \frac{158.5}{162} \times 100\% = 97.8\%.$$

(b)

$$\text{RTRR}_A = \frac{91}{35.48} = 2.55, \qquad \text{RTRR}_B = \frac{162}{158.5} = 1.02.$$

(c) The RTRR shows A to be preferable to B.

LESSON 9: BAYES' DECISION ANALYSIS

9.1 Introduction

In many cases the decision maker may need an expert's judgement to sharpen his or her assessment of the possible likelihood of a state of nature. Suppose that the consequences of an act depend on events which may be represented by the values of a random variable X with probability density function $p(x)$. Suppose further that the consequences are expressed in dollars of cost, so that the best act is that with the lowest expected cost. Consider the payoff table

Events	Probability	Act A	Act B
x	$p(x)$	$G_A(x)$	$G_B(x)$

with expected costs $E[G_A]$ and $E[G_B]$. We wish to examine whether another variable Y can be utilized to improve the decision. We assume that the distribution of Y does not depend on the act chosen.

9.2 Lesson learning outcomes

At the end of this lesson you will be able to determine posterior probabilities and determine the best decision under Bayes' decision-making approach.

9.2.1 Bayes' posterior probabilities

Let $p(x, y)$ be the joint distribution of X and Y, let $p(x)$ be the marginal distribution of X and $p(y)$ the marginal distribution of Y. Then the conditional distribution of X given Y is

$$P(X \mid Y) = \frac{P(X, Y)}{P(Y)}, \qquad P(Y) = \sum_x P(X, Y),$$

and the required conditional probabilities are given by Bayes' formula

$$P(X \mid Y) = \frac{P(X)\,P(Y \mid X)}{P(Y)}.$$

9.2.2 Bayes' decision-making approach

Example 1. A bond broker is considering a sale of her inventory holdings. Her profit on such a transaction would be $\$5,000$ if undertaken immediately. If she waits six months before selling, her profit will depend on the direction of change in interest rates during that period. The potential profits and the trader's subjective probabilities for each possible event are shown below.

State of nature	Probability	Profit
Lower interest rates	0.45	28,000
No change	0.35	$-5,000$
Higher interest rates	0.20	$-15,000$

Suppose that the latest business statistics show that a large number of companies are planning to raise capital through debt during the next six months. The probabilities of this situation, given expectations of lower rates, no change and higher rates, are 0.80, 0.20 and 0.10 respectively.

a. Compute the revised probabilities of the three events.

b. Determine the best decision using Bayesian decision analysis.

Solution.

(a) Prior expected profit. Before considering the additional information,

$$E(\text{wait}) = 0.45(28,000) + 0.35(-5,000) + 0.20(-15,000) = 12,600 - 1,750 - 3,000 = 7,850,$$

so her best action is to wait six months (compared with $\$5,000$ for selling now).

Revision table

State of nature	Prior $P(x)$	$P(\text{signal} \mid x)$	Joint	Posterior
Lower interest rates	0.45	0.80	0.36	0.800
No change	0.35	0.20	0.07	0.156
Higher interest rates	0.20	0.10	0.02	0.044
Total	1.00		**0.45**	**1.000**

> *[The module prints the prior expected profit of waiting as 7,800; its own three products give* $12,600 - 1,750 - 3,000 = 7,850$*. The decision is unchanged.]*

(b) Posterior expected profit.

$$E(\text{wait} \mid \text{signal}) = (-5,000)(0.156) + (28,000)(0.800) + (-15,000)(0.044) = \$20,960.$$

After considering the additional information her best action is still to wait six months.

Example 2. To save on expenses, Ron and Terry agreed to form a carpool for travelling to and from work in Nairobi City Centre. Ron preferred to use the somewhat longer but more consistent Juja road. Although Terry preferred the quicker Thika superhighway, she agreed with Ron that they should take Juja road if the Thika road had a traffic jam. The following payoff table provides the one-way time estimates in minutes for the trip to and from work.

Decision alternative	Thika road open, S1	Thika road jammed, S2
Juja road, D1	30	30
Thika road, D2	25	45

Based on their experience with traffic problems, Ron and Terry agreed on a 0.15 probability that the Thika road would be jammed. In addition, they agreed that the weather seemed to affect the traffic conditions on the Thika road.

a. Determine the optimal decision strategy of travel.

Solution.

State of nature	Probability	Juja, D1	Thika, D2	Expected D1	Expected D2
Thika road jammed, S2	0.15	30	45	4.5	6.75
Thika road open, S1	0.85	30	25	25.5	21.25
Total				**30**	**28**

> *[The module concludes "Optimal decision is to use Juja Road". Its own table gives an expected time of 30 minutes on the Juja road and 28 minutes on the Thika road, which favours the Thika road; the figures are reproduced as printed.]*

Activity 9.3 — Bayes' decision analysis

- **Title:** Bayes' decision analysis
- **Purpose:** To determine the best act using posterior probabilities.
- **Brief summary of overall task:** Watch *calculating Bayes' probability*.
- **Individual contribution:** Work out: from Example 2 above, complete part (b) — use Bayes' theorem for probability revision to determine the probability of each weather condition. Let C = clear, O = overcast and R = rain. The following conditional probabilities apply: $P\left(C \mid S_1\right) = 0.8$, $P\left(R \mid S_2\right) = 0.6$, $P\left(O \mid S_2\right) = 0.3$ and $P\left(R \mid S_1\right) = 0.0$.
- **Spark:** —
- **Dialogue begins:** Post your results on forum 9.
- **E-moderator interventions:** Giving feedback on students' work.
- **Time and schedule:** Week 9, for 3 hours.
- **Next:** Expected utility criterion.

9.4 Exercise

9.4.1 Vincent is a credit manager for the Fine Fabrics Limited (see Lesson 1). For companies whose actual credit record with the mill turns out to fall into each of the three categories, the following table shows the percentages that were given each of the three possible credit evaluations by the credit rating organizations.

Credit evaluation	Poor risk	Average risk	Good risk
Poor	50%	40%	20%
Average	40%	50%	40%
Good	10%	10%	40%

a. Develop a decision analysis formulation of this problem by identifying the decision alternatives, the states of nature and payoff table when the credit-rating organization is not used.

b. Assuming the credit rating organization is not used, use Bayes' decision rule to determine which decision alternative should be chosen.

Solution guide. (b) Bayes' rule gives an expected value of \$8, 000 for *extend credit* and \$0 for *do not extend credit*; the best decision is to extend credit.

> *[*$E(\text{extend}) = 0.2(-15,000) + 0.5(10,000) + 0.3(20,000) = 8,000$*, which agrees with the module's answer.]*

9.4.2 An investor has a certain amount of money available to invest. Three alternative investments are available, with estimated profits in dollars as indicated in the table below.

Event	A	B	C	Prior probabilities
Economy declines	500	$-2,000$	$-1,000$	0.2
No change	1,000	2,000	$-1,000$	0.5
Economy expands	2,000	5,000	20,000	0.3

Based on past experience the investor assigns the following probabilities to each economic condition: $P(\text{economy declines}) = 0.3$, $P(\text{no change}) = 0.5$ and $P(\text{economy expands}) = 0.2$.

Suppose that such focused returns could be predicted either for an expanding economy or for a declining or stagnant economy. Past experience indicates that with a declining economy, prior forecasts predicted an expanding economy 20% of the time; with no change of economy, prior forecasts predicted an expanding

economy 40% of the time; and when the economy expands, prior forecasts predicted an expanding economy 90% of the time. If the forecast is for an expanding economy, revise the given probabilities, hence find the best decision.

> *[The two sets of prior probabilities in this question disagree — the prose gives 0.3, 0.5 and 0.2 for decline, no change and expand, while the table's last column reads 0.2, 0.5 and 0.3. Both are reproduced; the prose values go with the sentence that follows them.]*

LESSON 10: EXPECTED UTILITY CRITERION

10.1 Introduction

The expected payoff or opportunity loss criterion sometimes fails to provide decisions that are consistent with the decision maker's attitude towards risk. In such cases we make use of the expected utility of each course of action.

10.2 Lesson learning outcomes

At the end of this lesson you will be able to formulate a utility function for a decision problem and determine the best decision using the expected utility criterion.

Example 1. Consider the following two actions:

- A1: Deposit your $\$100,000$ inheritance in the bank for one year at 7% interest.
- A2: Invest your $\$100,000$ inheritance for one year, with a 0.5 probability of having a total of $\$250,000$ at the end of the year and a 0.5 probability of losing your inheritance.

Determine the best decision.

Solution. A1 is most probably the best action if we base our decision on the guaranteed return, since the overall return will be $\$107,000$ at the end of the year, against a 50–50 chance of ending up with nothing. However, by choosing A1 the decision is not consistent with the action prescribed by the payoff criterion, since

$$E(A1) = 107,000(1) = 107,000, \qquad E(A2) = 250,000(0.5) = 125,000.$$

Using the payoff criterion, the best decision is A2.

The expected payoff criterion can be adapted to reflect our attitude towards risk if we can express the outcomes of the decision problem in terms of an objective variable that better reflects the true relative values of the outcomes. Such an objective variable is called a **utility function**, and the values assigned to the outcomes are called **utility values** or utilities.

Example 2. An individual lives in an area that has a probability of incurring cattle rustling, so he is considering buying insurance for his cattle at an annual cost of $\$180$. The probability of cattle rustlers coming for his livestock during one year is 0.001. If this happens, he estimates that the cost of damage — fully covered by the insurance policy — will be $\$160,000$. Suppose the total assets, including his home, are worth $\$250,000$, and suppose the utility function that measures how much he values his total assets X in dollars is

$$U(x) = \sqrt{x}.$$

Required: compare the utility of reducing the total assets next year by the cost of the cattle insurance policy with the expected utility next year of not taking the cattle insurance policy, and determine the best policy.

Solution.

Payoff table

Outcomes	Probability	Insured	Not insured
Cattle rustling occurs	0.001	249,820	90,000
Cattle rustling does not occur	0.999	249,820	250,000

Utility table, $U(x) = \sqrt{x}$

Outcomes	Probability	Insured	Not insured
Cattle rustling occurs	0.001	499.82	300.0
Cattle rustling does not occur	0.999	499.82	500.0

Expected utility table

Outcomes	Probability	Insured	Not insured
Cattle rustling occurs	0.001	0.49982	0.3
Cattle rustling does not occur	0.999	499.32	499.5
Expected utility		**499.819**	**499.8**

Since $499.819 > 499.8$,

$$U(\text{no insurance}) = 499.8 < 499.819,$$

and the optimal policy is to buy insurance.

Activity 10.3 — Expected utility criterion

- **Title:** Expected utility criterion
- **Purpose:** To determine the utility function and apply maximum utility in decision analysis.
- **Brief summary of overall task:** Watch *expected utility* and read on decision under uncertainty.
- **Individual contribution:** An oil company must make a decision concerning an option on a parcel of land. If the company takes the option it can drill with 100% interest, or with a 50% interest — that is, it splits all costs and profits with another firm. It costs $1,000,000 to operate a producing well until it is dry. The oil is worth $25 per barrel. Assume that the well is either dry or produces 200,000 or 400,000 barrels of oil. The firm assesses the probabilities of the outcomes as 0.8, 0.1 and 0.1 respectively.
  i. Based on the expected monetary return, determine the company's best action.
  ii. Suppose the company management has the following utility function:

Millions of dollars	−1.0	−0.5	1.5	3.0	4.0	8.0
Units of utility	−30	−15	45	90	120	240

- **Spark:** —
- **Dialogue begins:** Post your results above on the discussion forum 10.
- **E-moderator interventions:** Give feedback on discussions from the discussion forum; note any differences in your workings with those of your classmates; coordinate and direct discussion of learners, and improve on your workings.
- **Time and schedule:** Week 10, for 3 hours.
- **Next:** Decision trees.

10.4 Assessment

10.4.1 An individual with a net worth of $\$100,000$ contemplates investing in a risky venture with two equiprobable outcomes: it can either triple his net worth or bankrupt him completely. His preference for wealth W (net worth) can be represented by the utility function

$$U(W) = \sqrt{W}.$$

Determine how much he should be willing to pay at most if he is to maximize (i) the expected monetary return, (ii) the expected utility.

Solution guide. (i) $\$50,000$; (ii) expected utility $= 316.2$.

> *[With $U(W) = \sqrt{W}$ the gamble itself has expected utility $\frac{1}{2}\sqrt{300,000} + \frac{1}{2}\sqrt{0} = 273.9$, whose certain equivalent is $273.9^2 \approx 75,000$, so the most he would pay is about $\$25,000$; the printed value 316.2 is $\sqrt{100,000}$, i.e. the utility of his present worth. The module's answer is reproduced as printed.]*

LESSON 11: DECISION TREES

11.1 Introduction

In some cases the choice of an optimal act is not made in one stage, and the decision problem involves a sequence — not necessarily in time — of acts or events. There may be a number of basic alternatives leading to a number of situations depending on the outcome of a certain random process. At each such situation a number of other alternatives may be available, which also lead to a new set of situations depending on another set of events. The sequence of acts and events may be depicted in the form of a decision tree, and the decision problem is to find the most preferred branch of that tree.

11.2 Lesson learning outcomes

At the end of this lesson you will be able to apply a tree diagram to determine the best action in a decision problem.

Example. Joe's garage is considering hiring another mechanic. The mechanic would cost them an additional $\$50,000$ per year in salary and benefits. If there are a lot of accidents in Providence this year, they anticipate making an additional $\$75,000$ in net revenue. If there are not a lot of accidents, they could lose $\$20,000$ off last year's total net revenues. Because of anticipated "El Niño" rains, Joe thinks there will be a 70% chance of "a lot of accidents" and a 30% chance of "few accidents". Assume that if he does not expand, he will have the same revenue as last year. Using a decision tree, advise Joe on what to do.

Solution.

- *Start:* Hire new mechanic at $\$50,000$ **or** Don't hire at a cost of $\$0$.
- *If hire:* 70% chance of an increase in accidents with profit $\$70,000$, and 30% chance of a decrease with profit $-\$20,000$. The net present value of hiring a mechanic is
$$\text{NPV} = 0.7(70,000) + 0.3(-20,000) - 50,000 = 49,000 - 6,000 - 50,000 = -\$7,000.$$
- Therefore **do not hire the mechanic**.

> *[The introduction to the example quotes an additional revenue of* $\$75,000$*, while the solution uses* $\$70,000$*; with* $\$75,000$ *the NPV is* $0.7(75,000) + 0.3(-20,000) - 50,000 = -\$3,500$*. Both give the same advice.]*

Activity 11.3 — Decision making using decision trees

- **Title:** Decision trees
- **Purpose:** To determine a decision in stages.
- **Brief summary of overall task:** Watch *decision tree*; also watch *creating a decision tree*.
- **Individual contribution: (a)** Tom and Joe operate a rock quarry that provides local stone for landscaping. They currently have an offer of $50, 000 to sell the quarry. They are hesitant to sell because they believe there will be an increase in demand in the next 2 years that would improve their financial situation. If they decide to keep the quarry, the current buyer would be willing to pay $30, 000 in 2 years, regardless of the situation. Tom and Joe think there is a 60% chance of an increase, at which point they would operate at a profit of $75, 000 or sell the quarry to a new buyer for $60, 000. If demand does not increase, they would operate at a profit of only $10, 000. Use a decision tree to determine what they should do. **(b)** Vegetable Farms is a small, family-operated ranch that sells produce to local markets. The owners are currently trying to decide whether they should expand their operation next year. Because this is a fairly new business, the owners have assessed the following demand levels and probabilities: high (probability 0.50), medium (0.30) or low (0.20). The payoffs they expect for each demand/acreage scenario are listed below. Use a decision tree to help decide whether to expand the farm.

Acreage	High	Medium	Low
Same size	50,000	40,000	30,000
Expanded	100,000	40,000	−40, 000

- **Spark:** —
- **Dialogue begins:** Post your workings on forum 10.
- **E-moderator interventions:** Give feedback on discussions from the discussion forum; compare your workings with those posted by your classmates; coordinate and direct discussion of learners.
- **Time and schedule:** Week 11, for 3 hours.
- **Next:** Revision for the final examination.

11.4 Assessment

11.4.1 The athletics department of a local university is considering whether to hold an extensive campaign next year to raise funds for a new athletic field. The response to the campaign depends heavily upon the success of the football team this season. In the past, the football team has had winning seasons 60 percent of the time. If the football team has a winning season (W) this term, then many alumni will contribute and the campaign will raise $3 million. If the team has a losing season (L), few will contribute and the campaign will lose $2 million. If no campaign is undertaken, no costs are incurred. On September 1st, just before the football season begins, the athletics department needs to make its decision about whether to hold the campaign next year.

Suppose in similar situations in the past, when evaluating teams that have a winning season 50% of the time, James — a football guru — has had predictions that were correct 75% of the time. Considering that this team has more of a winning tradition, if James predicts a winning season:

Required: (a) Draw the decision tree for this entire problem. (b) Analyse this decision tree to determine the optimal policy regarding whether to undertake the campaign.

11.4.2 Decision making using trees. An electronic company manufactures high-fidelity equipment. Most of the components are made in-house, but some are purchased from other firms. One of these components is purchased from one supplier in lots of 10 items.

> *[The source page ends here (page 43 of 43); the remaining part of Question 11.4.2 is not present in the download.]*

APPENDIX A — REFERENCES

The module prints the following five references at the end of every lesson; they are collected here once instead of being repeated eleven times.

1. *Decision Making* (open access textbook with 121 solved exercises). ISBN 978-1979856959.
2. David, C. S., *Introduction to Decision Analysis* (3rd edition, 2018).
3. Abbas, Ali E. and Howard, Ronald A., *Foundations of Decision Analysis* (1st edition).
4. Sampson, D., *Managerial Decision Analysis* (Homewood, IL: Irwin, 1988).
5. Ulvila, J. W. and Brown, R. V., "Decision Analysis Comes of Age", *Harvard Business Review* (September 1982).

APPENDIX B — WHAT WAS BROKEN, AND WHAT WAS REPAIRED

B.1 The three text streams on every page

Each page of the downloaded file carried three independent streams of text laid on top of one another:

1. **the lesson prose**, set in the body column;
2. **the iLearn activity plan**, a two-column table whose cells are *Title, Purpose, Brief summary of overall task, Individual contribution, Spark, Watch, Dialogue begins, E-moderator interventions, Time and schedule, Next*; and
3. **the tables and figures**, whose cells had lost their grid, so that a table arrived as a single run of cell values.

Read straight out of the file, an activity therefore looked like this:

> "Brief summary of Watch decision making process Purpose DECISION MAKING PROCESS Dialogue begins Post your results for the above work out on the forum 1. Title overall task To develop a decision table"

In this edition each activity plan is set out as its own box, field by field, and every table is rebuilt as a table. The module plan of page 2, which arrived as "DECISION MAKING UNDER UNCERTAINTY WEEK 2 WEEK 5 OPPORTUNITY LOSS WEEK 9 BAYES' DECISION ANALYSIS …", is now the twelve-row table at the front of this document.

B.2 Tables rebuilt from their own arithmetic

The table values were recovered cell by cell and then checked against the arithmetic the module itself performs, so that a mis-transcribed cell would show up as an inconsistency. For example: the bonds row of Table 2.2.1 must be 12, 8, 7, 3 (not the "12 5 8 7" of the flattened text) because the module's own EMV line gives $12(0.4) + 8(0.3) + 7(0.2) + 3(0.1) = 8.9$ and its Hurwicz line uses max = 12, min = 3; and the regret table of Lesson 3 is exactly what that row produces.

B.3 Numbers that do not reconcile in the source

Each of these is printed as it stands in the text, with the corrected value in a note at that point.

#	Page	As printed	Recomputed
1	4	Activity 1.3 "Post your results … on the forum 1" (the field order is jumbled: *Title* and *Purpose* values sit under the wrong labels)	fields assigned by meaning
2	13	Bonds row of the Lesson 4 payoff table: 12, 8, 8, 3	12, 8, 7, 3 (the module's own EMV of 8.9)
3	14	EMV of stocking 14 bags: 8.6	8.0
4	11	Regret table for the (cost) staffing problem computed as max − cost	cost − min gives 50/600/300, 50/300/150, 0/0/0 and the minimax act is *combination*
5	16	"Post your workings on forum 3" in Lesson 4	forum 4
6	21	Expected payoff under certainty 460,000	460,500 (its own three products; needed for the stated EVPI of 5,500)
7	21	Gears & levers expected profit 421,500	412,500 (as printed in Lesson 7)
8	26	Office A3 expected opportunity loss 65,000	68,000 (25,000 + 0 + 43,000)
9	29	"RTRR … gradual introduction has much higher risk than the concentrated strategy"	gradual has the higher RTRR (1.497 against 0.136) and the lower CV (66.8% against 734.8%), which is what the module's conclusion uses
10	33	Carpool: "Optimal decision is to use Juja Road"	the table's own expectations are 30 minutes (Juja) and 28 minutes (Thika)

#	Page	As printed	Recomputed
11	34	$P(R \mid S_2) = 0.6$, $P(O \mid S_2) = 0.3$	these two exceed 1 together with $P(C \mid S_2)$; reproduced as printed
12	36	Prior probabilities 0.3, 0.5, 0.2 in the prose but 0.2, 0.5, 0.3 in the table	both reproduced
13	40	Expected utility 316.2 for $U(W) = \sqrt{W}$	the gamble's expected utility is 273.9 (certain equivalent ≈ 75,000)
14	41	Joe's garage: revenue $75,000 in the question, $70,000 in the solution	NPV −3,500 with 75,000, −7,000 with 70,000; advice unchanged
15	17	Subsections numbered 5.5.1 and 5.5.2	renumbered 5.2.1 and 5.2.2 (they follow §5.2)

B.4 Figures

The diagrams of the module (the week-by-week module plan, the payoff and loss tables, the decision trees of Lesson 11, and the layout grids of the activity boxes) exist in the download only as empty ruled boxes — the lines are drawn, the labels are text laid over them. The tables have therefore been rebuilt as tables; the decision trees are given as the staged text of the worked solutions, since no tree image survives in the file.

B.5 Course code

The title page reads **SST 402: Decision Analysis**; the file name supplied reads *SST 401*. Both are noted; the document keeps the title page's code.

APPENDIX C — SOURCE PAGE MAP

The source has 43 pages; all are represented in this edition.

Page	Content
1	Title page: Kenyatta University, Digital School of Virtual and Open Learning, SST 402 Decision Analysis, author
2	Module plan (weeks 1–13); Lesson 1 introduction and learning outcomes
3	§1.2.1 decision-making environment; Vincent credit example; Table 1.1
4	Payoff table of the Vincent example; Activity 1.3
5	§1.4 Exercise 1.4.1 and its solution table; Lesson 2 introduction
6	§2.1 optimism/pessimism; §2.2 outcomes; Table 2.2.1; pessimism solution
7	Optimism and Hurwicz solutions; Activity 2.3
8	Activity 2.3 table; §2.4 assessment questions
9	Solutions to 2.4.1; Question 2.4.2; Lesson 3 introduction
10	§3.2 steps; regret table (Table 3.2.1); Activity 3.3
11	Activity 3.3 question (staffing costs); §3.4 assessment
12	Staffing regret table; Question 3.4.2; Lesson 4 introduction
13	§4.2.1–4.2.2 steps; Example 1 (EMV); Example 2 begins
14	Example 2 payoff and expected payoff tables; Activity 4.3
15	Activity 4.3 case analysis (Special Occasions)
16	Demand table; §4.4 questions 4.4.1 and 4.4.2
17	Costs of 4.4.2; Lesson 5 introduction and outcomes
18	Table 5, loss table, expected loss; Activity 5.3
19	Activity 5.3 question (toy advertising); §5.4 Question 5.4.1 (garment)
20	5.4.1 payoff and loss tables; Lesson 6 introduction
21	§6.2 steps; EVPI example with EPPI tables
22	EVPI interpretation; Activity 6.3; §6.4 Question 6.4.1
23	6.4 solution: conditional profit and expected profit tables
24	EVPI = Ksh 101; Lesson 7 introduction
25	§7.2 relationship and Table 7.2.1; Activity 7.3
26	§7.4 Questions 7.4.1 (real estate) and 7.4.2 (author)
27	Lesson 8 introduction and outcomes
28	§8.2.1 variance, CV; salad-dressing example: payoff and EMV tables
29	CV and RTRR computations; Activity 8.3
30	Activity 8.3 question; §8.4 assessment (stocks A and B)
31	8.4 solution (CV and RTRR); Lesson 9 introduction
32	§9.2.1 Bayes' posterior probabilities; Example 1 (bond broker)
33	Example 1 revision table and posterior EMV; Example 2 (carpool)
34	Example 2 solution table; Activity 9.3
35	§9.4 Question 9.4.1 (Vincent)
36	Question 9.4.2 (investor forecasts); Lesson 10 introduction
37	§10.2 Examples 1 and 2 (bank deposit; cattle rustling)
38	Example 2 payoff/utility/expected-utility tables; Activity 10.3
39	Activity 10.3 question (oil company) and utility function; §10.4

Page	Content
40	Question 10.4.1 and answer; Lesson 11 introduction
41	Joe’s garage example; Activity 11.3
42	Activity 11.3 (a) and (b) with the acreage table; Question 11.4.1
43	11.4.1 continued; Question 11.4.2 (begins, and is cut off in the source)